

■■ PathFundAI Technology & Data Overview

1■■ Career Pathway Generation

- **AI:** Google Gemini 1.5 Pro for reasoning and structured JSON output.
- **Data Sources:** MVP uses a curated JSON seed of MDC programs; stretch goal includes parsing official MDC catalog pages or PDFs.
- **Backend:** Cloudflare Workers + KV/D1 storage.
- **Goal:** Combine official MDC data for accuracy with AI reasoning for pacing and alternatives.

2■■ Scholarships (Discovery & Matching)

- **MVP:** Curated JSON of 10–20 scholarships (award, eligibility, essay length, URL).
- **Stretch:** Fetch structured data from whitelisted scholarship pages using Cloudflare Workers.
- **AI:** Gemini filters scholarships relevant to student goals.
- **Matching:** Keyword or embeddings-based ranking (Cloudflare Workers AI).
- **Input RFP:** Allow user to paste scholarship text or upload a PDF as the Request for Proposal.

3■■ ENTEC Proposal Generation

- **Input:** Scholarship description or uploaded RFP PDF.
- **Template:** Structured sections (Executive Summary, Objectives, Budget, Impact).
- **AI:** Gemini extracts requirements and fills the defined template with user pathway and NFT data.
- **Output:** PDF/DOCX via Markdown-to-PDF conversion.
- **Goal:** Fully satisfy ENTEC's requirement to generate a proposal from a given RFP and template.

4■■ Solana NFT Credentials

- **Wallet:** Phantom (Devnet).
- **Libraries:** @solana/web3.js, Metaplex JS.
- **Metadata:** milestone ID, user PK, timestamp, optional evidence hash.
- **Purpose:** Mint Proof-of-Completion NFTs for verified milestones.

5■■ Infrastructure & Frontend

- **Frontend:** React + Tailwind + shadcn/ui (optional).
- **Backend:** Cloudflare Workers with Hono routing.
- **AI Integration:** Gemini REST API with Zod validation.
- **Storage:** Cloudflare KV for caching, D1 for persistent user/milestone data.
- **Deployment:** Cloudflare Pages + Wrangler CLI.

6■■ Data Acquisition Strategy

- **Scholarships:** Begin with curated JSON; optionally add “Paste RFP” input for flexibility.
- **MDC Programs:** Reference official MDC documents as the truth source, storing URLs and timestamps.
- **AI Augmentation:** Gemini transforms verified data into paced, alternative roadmaps.
- **Approach:** Combine official data for accuracy with AI for personalization.

■ MVP vs Stretch Goals

MVP: Curated MDC + Scholarship JSON, Gemini for pathway and proposal generation, Solana Devnet NFT mint, React UI.

Stretch: Automated catalog/scholarship parsing, embeddings ranking, IPFS metadata storage, enhanced dashboard.